

D892

Multivariable Calculus

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A game designer at a studio with 45 employees checks whether two direction vectors $u = (2, -1, 3)$ and $v = (1, 5, 1)$ meet at a right angle. The studio's current game is set on a space station. What is true?

- A. You cannot tell without their magnitudes.
- B. They are parallel, because their dot product is 0.
- C. They meet at 45 degrees, because the components have mixed signs.
- D. They are orthogonal, because their dot product is 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

For what value of k are $(k, 2, -1)$ and $(3, k, 4)$ orthogonal?

- A. $k = 4/5$
- B. $k = -4/5$
- C. $k = 4/3$
- D. $k = 4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A structural engineer at a firm that designs stadiums finds the area of a flat panel spanned by $u = (2, 0, 1)$ and $v = (0, 3, 1)$ meters. The panel will be painted blue. What is the area of the parallelogram?

- A. 3.5 square meters
- B. 7 square meters
- C. 49 square meters
- D. 1 square meter

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

For two nonzero vectors, $u \times v = (0, 0, 0)$. What must be true?

- A. One of the vectors must be the zero vector.
- B. The vectors have equal length.
- C. The vectors are parallel.
- D. The vectors are orthogonal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A heat model for a new classroom building is $f(x, y) = x^2 y^3 + 4xy$. The building has solar panels on the roof. What is f_x at $(2, 1)$?

- A. 12
- B. 20
- C. 8
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hiker's elevation on a trail map sold at the park gift shop is modeled by $f(x, y) = x^2 + 3y^2$. The map is waterproof. At $(1, 1)$, in which direction does elevation increase fastest?

- A. In the direction of $(-2, -6)$
- B. In the direction of $(6, 2)$
- C. In the direction of $(1, 1)$
- D. In the direction of $(2, 6)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

For $f(x, y) = x^2 + 3y^2$ at $(1, 1)$, what is the maximum rate of increase?

- A. 8
- B. About 6.32
- C. 40
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An optimization problem for a shipping company that runs 60 trucks finds a critical point where $f_{xx} = 4$, $f_{yy} = 2$, and $f_{xy} = 3$. The company is headquartered near a port. What kind of point is it?

- A. A saddle point
- B. A local maximum
- C. A local minimum, because $f_{xx} > 0$.
- D. The test is inconclusive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A tutor preparing notes in a coffee shop rewrites the integral from $x = 0$ to 2 of the integral from $y = 0$ to x of $f(x, y) dy dx$ with dx on the inside. The notes are for a Saturday review session. Which is correct?

- A. The integral from $y = 0$ to 2 of the integral from $x = y$ to 2 of $f dx dy$
- B. The integral from $y = 0$ to x of the integral from $x = 0$ to 2 of $f dx dy$
- C. The integral from $y = 0$ to 2 of the integral from $x = 0$ to 2 of $f dx dy$
- D. The integral from $y = 0$ to 2 of the integral from $x = 0$ to y of $f dx dy$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A drone at a delivery startup flies a winding path from $(0, 0)$ to $(2, 3)$ through the field $F = (2xy, x^2)$. The route is 4.7 km long and passes over a river. What is the work done by F ?

- A. It depends on the path.
- B. 12
- C. 4.7 times 12
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

$F = (2xy, x^2)$. What is the line integral of F around a closed circle?

- A. 12
- B. 2 pi
- C. It depends on the circle's radius.
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Engineers at a lab that tests weather balloons compute the outward flux of $F = (x, y, z)$ through a sphere of radius 2. The lab is located at high altitude. What is the flux?

- A. 0
- B. $32\pi / 3$
- C. 32π
- D. 16π

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A game designer ~~at a studio with 45 employees~~ checks whether two direction vectors $u = (2, -1, 3)$ and $v = (1, 5, 1)$ meet at a right angle. ~~The studio's current game is set on a space station.~~ What is true?

Target word: angle between vectors **Decides:** $u \cdot v = 2 - 5 + 3 = 0$.

A	Dropped element	The dot product alone settles orthogonality.
B	Confusion set	Parallel vectors have a cross product of 0, not a dot product of 0.
C	Adjacent category	Component signs do not give the angle.
D	Correct	A zero dot product means a 90 degree angle.

2. Answer A Concept 1 · transfer

For what value of k are $(k, 2, -1)$ and $(3, k, 4)$ orthogonal?

Target word: solve for k **Decides:** Set the dot product to 0: $3k + 2k - 4 = 0$.

A	Correct	$5k = 4$.
B	Inverted operation	Sign error moving the 4.
C	Dropped element	Drops the $2k$ term.
D	Wrong stem number	Forgets to divide by 5.

3. Answer B Concept 2

A structural engineer ~~at a firm that designs stadiums~~ finds the area of a flat panel spanned by $u = (2, 0, 1)$ and $v = (0, 3, 1)$ meters. ~~The panel will be painted blue.~~ What is the area of the parallelogram?

Target word: parallelogram area **Decides:** $|u \times v|$ with $u \times v = (-3, -2, 6)$.

A	Confusion set	That is the triangle, half the parallelogram.
B	Correct	The square root of $9 + 4 + 36 = 7$.
C	Dropped element	Forgets the square root.
D	Confusion set	That is the dot product, which does not give area.

4. Answer C Concept 2 · transfer

For two nonzero vectors, $u \times v = (0, 0, 0)$. What must be true?

Target word: meaning of zero cross product **Decides:** $\sin \theta = 0$.

A	Absolute language	The stem says both are nonzero; parallel vectors also give 0.
B	Adjacent category	Length does not make the cross product zero.
C	Correct	A zero cross product means zero area, so the angle is 0 or 180 degrees.
D	Confusion set	Orthogonality makes the dot product 0.

5. Answer C Concept 3

A heat model for a new classroom building is $f(x, y) = x^2 y^3 + 4xy$. ~~The building has solar panels on the roof.~~ What is f_x at $(2, 1)$?

Target word: f_x at a point **Decides:** $f_x = 2xy^3 + 4y$.

A	Adjacent category	That is $f(2, 1)$, the function value.
B	Confusion set	That is $f_y = 3x^2 y^2 + 4x$ at the point.
C	Correct	$2(2)(1) + 4(1) = 8$.
D	Dropped element	Drops the $4y$ term.

6. Answer D Concept 4

A hiker's elevation on a trail map ~~sold at the park gift shop~~ is modeled by $f(x, y) = x^2 + 3y^2$. ~~The map is waterproof.~~ At $(1, 1)$, in which direction does elevation increase fastest?

Target word: direction of steepest ascent **Decides:** $\text{grad } f = (2x, 6y) = (2, 6)$.

A	Inverted operation	That is steepest descent.
B	Confusion set	Swaps the partial derivatives.
C	Confusion set	That is the position vector of the point, not the gradient.
D	Correct	The gradient points uphill fastest.

7. Answer B Concept 4 · transfer

For $f(x, y) = x^2 + 3y^2$ at $(1, 1)$, what is the maximum rate of increase?

Target word: max directional derivative **Decides:** $|\text{grad } f| = \text{square root of } 40$.

A	Adjacent category	Adds the partials instead of taking the length.
B	Correct	The square root of $4 + 36 = \text{square root of } 40$.
C	Dropped element	Forgets the square root.
D	Confusion set	That is the rate along a level curve.

8. Answer A Concept 5

An optimization problem for a shipping company ~~that runs 60 trucks~~ finds a critical point where $f_{xx} = 4$, $f_{yy} = 2$, and $f_{xy} = 3$. ~~The company is headquartered near a port.~~ What kind of point is it?

Target word: classify the point **Decides:** $D = 8 - 9 = -1 < 0$.

A	Correct	$D < 0$.
B	Inverted operation	A max needs $D > 0$ and $f_{xx} < 0$.
C	Dropped element	Ignores the mixed partial; D must be positive first.
D	Confusion set	Inconclusive only when $D = 0$.

9. Answer A Concept 6

A tutor ~~preparing notes in a coffee shop~~ rewrites the integral from $x = 0$ to 2 of the integral from $y = 0$ to x of $f(x, y) dy dx$ with dx on the inside. ~~The notes are for a Saturday review session.~~ Which is correct?

Target word: reversed limits **Decides:** The region is the triangle $0 \leq y \leq x \leq 2$.

A	Correct	For each y , x runs from the line $y = x$ to $x = 2$.
B	Dropped element	Just swaps the limits without redrawing; the outer limit cannot contain x .
C	Absolute language	That is the whole square, not the triangle.
D	Inverted operation	That describes the other triangle, above the line $y = x$.

10. Answer B Concept 7

A drone at a delivery startup flies a winding path from $(0, 0)$ to $(2, 3)$ through the field $F = (2xy, x^2)$. ~~The route is 4.7 km long and passes over a river.~~ What is the work done by F ?

Target word: line integral of a conservative field **Decides:** $F = \text{grad}(x^2 y)$.

A	Confusion set	A gradient field is path independent.
B	Correct	$f(2, 3) - f(0, 0) = 4 \times 3 - 0 = 12$.
C	Camouflage	The path length does not change the result.
D	Absolute language	Zero only for a closed loop; this path has different ends.

11. Answer D Concept 7 · transfer

$F = (2xy, x^2)$. What is the line integral of F around a closed circle?

Target word: closed-loop value **Decides:** Conservative field.

A	Wrong stem number	Uses the endpoint values from a different path.
B	Adjacent category	That is the value for some non-conservative fields around the unit circle.
C	Confusion set	Path independence means radius does not matter.
D	Correct	The start and end points are the same, so $f(B) - f(A) = 0$.

12. Answer C Concept 8

Engineers ~~at a lab that tests weather balloons~~ compute the outward flux of $F = (x, y, z)$ through a sphere of radius 2 . ~~The lab is located at high altitude.~~ What is the flux?

Target word: flux by divergence theorem **Decides:** $\text{div } F = 3$ times the volume.

A	Absolute language	Only a divergence-free field gives zero flux here.
B	Dropped element	Drops the divergence value 3 .
C	Correct	$3 \times (4/3) \pi (2^3) = 32 \pi$.
D	Confusion set	That is the sphere's surface area.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Dot product and orthogonality	$u \cdot v = u v \cos \theta$; two nonzero vectors are orthogonal exactly when their dot product is 0.	Thinking a zero dot product means parallel, or that magnitudes are needed to test for a right angle.	Two vectors in component form and a question about the angle between them.
2	Cross product magnitude and direction	$ u \times v $ is the area of the parallelogram spanned by u and v (half of it for the triangle), and $u \times v$ is perpendicular to both; $u \times v = 0$ for parallel vectors.	Using the dot product for area, or forgetting the triangle is half.	An area in space or a normal vector.
3	Partial derivatives	To find f_x , treat every other variable as a constant and differentiate with respect to x .	Differentiating with respect to all variables at once, or mixing up f_x and f_y .	A function of two variables and a request for one partial derivative at a point.
4	The gradient and steepest ascent	The gradient points in the direction of fastest increase, and its length is the maximum rate of change; it is perpendicular to level curves.	Thinking the fastest rate is the sum of the partials, or that the gradient points downhill.	Which direction a quantity increases fastest, or the maximum directional derivative.
5	Second derivative test in two variables	At a critical point, $D = f_{xx}f_{yy} - (f_{xy})^2$: $D > 0$ with $f_{xx} > 0$ is a min, $D > 0$ with $f_{xx} < 0$ is a max, $D < 0$ is a saddle, $D = 0$ is inconclusive.	Using only the sign of f_{xx} and ignoring the mixed partial.	Second partials given at a critical point.
6	Changing the order of integration	Rewrite the same region: sketch it, then describe it with the other variable on the outside, with inner limits as functions of the outer variable.	Swapping the limits without redrawing the region.	A double integral with a variable limit and a request to switch dx and dy .
7	Conservative fields and path independence	If $F = \text{grad } f$, the line integral from A to B equals $f(B) - f(A)$ for any path, and it is 0 around any closed loop.	Computing a long path integral term by term, or thinking the path shape changes the answer.	A field that is a gradient, or a curl of zero on a simply connected region.
8	Divergence theorem	Outward flux through a closed surface equals the triple integral of $\text{div } F$ over the enclosed solid.	Using surface area instead of volume, or dropping the divergence value.	A closed surface such as a sphere or box, and flux asked.

Course scope (public description): Multivariable Calculus provides an advanced exploration of calculus in multiple dimensions. The course begins with vector operations in both two and three dimensions, with emphasis on mastering computations and interpretations of magnitudes, directions, and vector products. Then, it presents partial differentiation and the integration of functions of multiple variables. Finally, the course teaches learners how to analyze applications of vector fields using line, surface, and volume integrals. This course integrates and extends previous calculus concepts, providing students with the opportunity to gain a deep understanding of key mathematical theorems while preparing students for specialized courses and professional applications. Calculus II is a prerequisite for this course.